

Can We Trust Gaze? A Reliability and Fairness Study of Webcam Gaze Estimation in Online Learning

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Abstract—Online learning platforms increasingly estimate student engagement from webcam gaze, often assuming that the underlying gaze signal is reliable for every student. However, this assumption has received little scrutiny. We present the Gaze Reliability Index (GRI), a no-reference measure that scores the trustworthiness of webcam gaze extraction directly from the signal itself. We evaluate GRI on 8,409 confirmed high-engagement clips of 112 students from the DAiSEE dataset. We find that gaze extraction never visibly fails, i.e., detection succeeds in 99.6% of clips. Yet behind this apparent reliability, signal quality varies substantially, ranging from 0.35 to 0.95 on the GRI scale. Additionally, 71% of this variation is explained by student identity alone, indicating that the same students receive degraded signal across sessions. Low GRI also biases human annotation: students with poorer signal are labelled less engaged and receive more internally conflicting labels, turning a measurement disadvantage into a judgment disadvantage. Blinded human raters independently confirm that GRI reflects visible gaze quality (Spearman $\rho = 0.70$). The dominant drivers are person-stable eye-region darkness, iris image quality, and temporal instability, while eyeglasses, the commonly assumed culprit, show no meaningful effect in our data. We argue that gaze-based educational systems should treat signal reliability as a transparent, per-student measure, and GRI provides an audit instrument for this at no labelling cost.

Index Terms—Webcam gaze estimation, gaze reliability, eye tracking, algorithmic fairness, student engagement, learning systems, online learning.

I. INTRODUCTION

Webcam-based gaze estimation has become a primary signal for inferring student attention in online learning systems. DAiSEE [1], with 9,068 ten-second clips from 112 students across six naturalistic settings and three illumination levels, is a standard benchmark for engagement recognition from webcam video; systems built on it monitor attention, detect disengagement [2]. These systems implicitly assume that gaze-derived signals can be extracted reliably across students and recording conditions.

Recent engagement studies directly use gaze signals extracted through LSTM ensembles [3], MediaPipe-based landmark detection with graph neural networks [4], and multi-modal gaze-posture fusion [5], without explicitly evaluating whether gaze extraction reliability varies across subjects. To the best of our knowledge, prior engagement detection research has not systematically examined gaze reliability as a subject-level property within an engagement benchmark.

Das and Dev [2] report that gaze and head pose features underperform facial action units on WACV, but it remains unclear whether this reflects weaker predictive power or unreliable feature extraction. Likewise, Almuniri et al. [6] show substantial variation in model stability across feature configurations, highlighting the need to evaluate feature reliability directly rather than infer it from downstream accuracy. Good prediction performance does not guarantee reliable signal extraction. Therefore, the reliability of gaze extraction itself remains an open question, even though unequal signal quality may introduce measurement bias before any engagement model is trained or evaluated.

This gap is not merely a technical inconvenience. If gaze failures cluster around students with physical traits they cannot change, such as naturally narrow eye openness or a medical need for corrective eyewear, any gaze-based system may be structurally unfair to those students regardless of aggregate accuracy. Buolamwini and Gebru [7] established that appearance-based systems fail unequally across groups, with disparities tied to underrepresentation in training data; Yin et al. [8] document the same pattern in educational AI, where bias disproportionately affects marginalised learners. Akgül et al. [9] conduct a systematic fairness evaluation of appearance-based gaze estimation, finding significant disparities that current systems do not address. Understanding who is affected and why is essential for equitable intelligent educational systems.

This paper addresses the following overarching question:

does confirmed student engagement guarantee reliable webcam gaze extraction across student populations? Three research questions follow. **RQ1:** How does gaze reliability vary among confirmed high engagement students, and which factors drive low reliability? **RQ2:** Are gaze reliability failures randomly distributed across students, or do they systematically cluster around specific student characteristics? **RQ3:** Does the proposed Gaze Reliability Index capture meaningful variation in gaze signal quality that reflects real differences in signal trustworthiness?

The contributions of this paper are as follows:

- 1) **GRI (Gaze Reliability Index):** The first no-reference gaze quality index for webcam educational video, integrating five complementary reliability dimensions into a single interpretable score.
- 2) **First large-scale reliability audit:** The first large-scale audit of webcam gaze reliability in educational AI, quantifying signal quality independently of engagement-prediction performance and demonstrated on the full DAiSEE corpus.
- 3) **Fairness finding:** Evidence that these failures are not randomly distributed but cluster around specific students, raising potential fairness concerns for gaze-based educational AI systems.

The remainder of this paper is structured as follows. Section 2 reviews related work across gaze estimation, engagement detection, and quality as well as fairness literature. Section 3 presents the GRI framework, experimental design, and human validation. Section 4 reports the results. Section 5 discusses reliability and fairness implications and outlines limitations and future work. Section 6 concludes.

II. RELATED WORK

This paper sits at the intersection of three research areas: webcam gaze estimation under real-world conditions, gaze-based student engagement detection, and signal quality and fairness analysis in AI systems. The following review highlights how reliability remains largely overlooked across these areas and motivates the proposed Gaze Reliability Index (GRI).

A. Webcam Gaze Estimation

Appearance-based webcam gaze estimation has improved with large annotated datasets and deep learning. Wood *et al.* [10] synthesize photo-realistic, automatically labelled eye images across varied head poses, gaze directions, and lighting, showing that illumination is critical: accuracy drops sharply when test lighting differs from training.

Zhang *et al.* [11] introduced MPIIGaze and the first deep appearance-based method, GazeNet, identifying gaze range, illumination, and personal appearance as the main degradation factors, with the worst accuracy for glasses-wearers, an assumption that this study tests rather than inherits.

Later datasets widened the range of poses and conditions. Gaze360 [12] outputs a per-prediction uncertainty because accuracy degrades when the eye is seen from the side or

hidden, and ETH-XGaze [13] confirms that poor lighting and large pose or gaze angles measurably reduce accuracy.

Falch and Lohan [14] project webcam gaze onto the screen and compare it with dedicated hardware, finding webcams sensitive to head movement and lighting, since reflections on the eye distort the estimate, and still less accurate than purpose-built trackers.

However, existing work focuses primarily on prediction accuracy rather than signal reliability. While gaze-estimation errors are routinely reported, little attention has been paid to whether gaze can be extracted first with comparable reliability across users and recording conditions. This gap between reported accuracy and unexamined extractability motivates the present work.

B. Gaze for Student Engagement Detection

Gaze has become a key feature in automated student engagement detection. Much of this work builds on the DAiSEE benchmark [1].

Buono *et al.* [3] predict engagement intensity with an ensemble of three Long Short-Term Memory networks over separate eye-gaze, head-pose, and action-unit streams. Although gaze is treated as a first-class input, their analysis found stronger associations between engagement and facial action units and head-pose features than gaze behaviour.

Abedi and Khan [4] use gaze implicitly, via a spatial-temporal graph network over MediaPipe landmarks encoding head pose, action units, and gaze; among single features, eye gaze is consistently the weakest.

Stidi and Nasser [5] fuse MediaPipe FaceMesh gaze angles with body-posture keypoints for engagement recognition under occlusion; their gaze-only models were often unable to predict a confident engagement state when faces are occluded, an observation directly relevant to reliability, although treated only as a motivation for fusion.

Das and Dev [2] compare feature sets, namely facial landmarks, eye gaze with head pose, action units, and their combination; the gaze-and-head-pose set is again the weakest (e.g., 0.80 versus 0.82–0.83 with XGBoost), though gaze is not isolated from head pose and the ranking is on WACV, not DAiSEE.

Across these systems, gaze is used either as an explicit input or as part of a landmark representation, and several studies report that it is the weakest cue. In every case, however, it is passed to the classifier without checking whether it can be extracted reliably from a clip. Extraction failure is at most noted qualitatively [5] or absorbed into aggregate accuracy [2], [4]; it is never measured per clip, nor treated as varying systematically across students. This unverified assumption, that usable gaze can be obtained from every learner, is the gap the present work addresses.

C. Signal Quality Metrics and Fairness in Vision Systems

Reliability and fairness concerns in vision systems have been studied independently in two separate bodies of literature.

The first concerns signal quality: judging how usable a measurement is without a clean reference. Mittal *et al.* introduced BRISQUE [15], a no-reference model that quantifies how far an image’s locally normalized luminance statistics deviate from natural ones, requiring neither a reference nor a known distortion. It established the no-reference philosophy adopted by GRI; since BRISQUE was tuned on synthetic distortions and yields one holistic score, GRI inherits the philosophy rather than the metric.

Jakobi *et al.* [16] argue eye-tracking data quality must be measured and reported per recording through standardized metrics (data-loss ratio, calibration and validation scores, blink ratios), motivating GRI’s per-clip reporting, though their target is infrared trackers rather than webcams. Pessaud *et al.* [17] showed that eyelid occlusion and pupil dilation degrade iris recognition and that the visible iris area, computed from a single image prior to matching, correlates strongly with matching error ($r = -0.774$), paralleling GRI’s iris image-quality and iris-detection components as no-reference signals indicating whether a sufficient portion of the eye is visible.

The second body of literature regards demographic parity, where the consistent result is that differences in accuracy are systematic rather than random. Buolamwini and Gebru [7] evaluated three commercial gender-classification systems in their Gender Shades audit, revealing error rates up to 34.7% for darker-skinned women versus 0.8% for lighter-skinned men. The authors attribute the gap to under-representation in training data rather than skin tone itself, a distinction that matters for GRI, which treats observability rather than group identity as the measurable quantity.

Yin *et al.* [8] document the same biases in educational AI, but for predictive systems such as grading, dropout prediction, and admissions, not whether a sensor signal can be obtained at all. Most directly, Akgül *et al.* [9] present a systematic fairness study of appearance-based gaze estimation, finding significant disparities that standard bias-mitigation barely addresses because it targets data quantity while leaving the observability of the gaze signal across groups untouched.

These two literatures have remained separate, and neither has been applied to webcam gaze estimation in education: quality metrics target generic images, specialized eye trackers, or iris biometrics, while gaze-fairness work targets controlled benchmarks rather than the failure of extraction itself.

Across all three threads, one assumption goes unexamined: that gaze can be reliably recovered from every learner; webcam estimation degrades under precisely the conditions common in real classrooms (Section II-A), engagement systems build on these estimates without verifying that the base signal exists (Section II-B), and the quality and fairness literature shows usability can be measured reference-free and that failures may vary systematically across users, not at random. GRI sits at this intersection, turning the implicit assumption of reliable gaze extraction into an explicit per-clip measure that we define and evaluate on the DAiSEE dataset in the remainder of this paper.

III. METHODOLOGY

This section presents our framework for measuring webcam gaze reliability. Figure 1 shows the overall pipeline. We first describe the dataset (III-A), then the extraction pipeline and how we validated it (III-B), the Gaze Reliability Index and its components (III-C), the failure taxonomy (III-D), the statistical analyses that address RQ1 to RQ3 (III-E), and a blinded human validation of the index (III-F).

Study design. This is an observational audit, not a model benchmark. We hold the behavioural ground truth fixed by analysing only clips with crowd-validated high engagement, so that any reliability deficit we measure cannot be attributed to the student looking away out of boredom. On this controlled population we measure signal reliability per clip with a no-reference index, then test three hypotheses: that degradation exists (RQ1), that it is systematic per student rather than random (RQ2), and that it predicts the trustworthiness of human labels (RQ3).

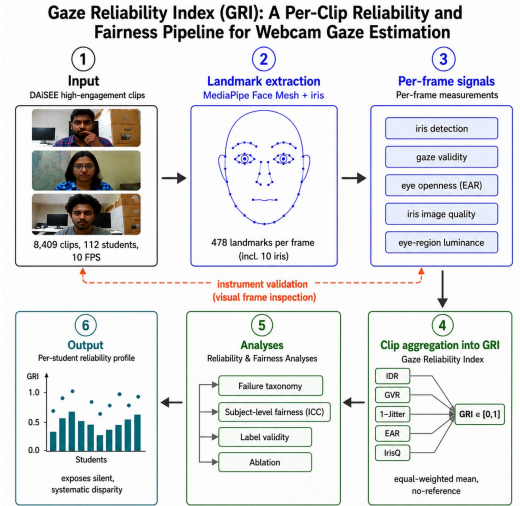


Fig. 1. Overview of the GRI pipeline. Frames are sampled at 10 FPS, facial and iris landmarks are extracted per frame, five per-frame signals are aggregated into clip-level components, and the components are combined into a single reliability score that feeds the taxonomy, fairness, and validation analyses.

A. Dataset

We use DAiSEE [1], the largest public in-the-wild video dataset of student affect. It contains recordings of 112 students in natural e-learning settings, with crowd-validated labels for engagement, boredom, confusion, and frustration. The public copy used in this study contains 8,925 ten-second clips at 30 fps (a publicly redistributed copy; the 143 clips absent relative to the original release of 9,068 are not used in any analysis). We restrict the analysis to the 8,409 clips labelled High (4,422) or Very High (3,987) engagement. This restriction is the core of our design. Any reliability failure found on these clips cannot be explained by a student looking away out of boredom. It must come from student characteristics or recording conditions. Eighteen subjects wear eyeglasses, identified by

visual inspection of sample frames (two tinted, ten thick-framed, six thin-framed). The dataset ships no illumination labels, so illumination is measured directly from eye-region brightness as described below. DAiSEE was collected with informed consent, and this study analyses signal reliability in aggregate to make gaze-based systems more equitable for the students they observe.

B. Extraction Pipeline and Validation Protocol

Each clip is sampled at 10 FPS with OpenCV, giving about 100 frames per clip. Every frame is processed with MediaPipe Face Mesh [18], [19] with iris refinement enabled, which returns 478 facial landmarks, including ten iris landmarks [20]. From the landmarks we compute six per-frame signals: iris detection, geometric gaze validity, eye openness, iris-centred image quality, eye-region luminance (used as a diagnostic), and the iris centre in pixel coordinates (used to measure temporal stability). Processing is checkpointed and reproducible, with a fixed library version and random seed.

Every measurement was checked against direct visual inspection before adoption, which corrected three instruments. A single-pair eye aspect ratio under-reads openness, so we use the two-pair form. Whole-frame brightness misleads for webcams, since a bright background drives underexposure of the face (frame and eye-region brightness correlate negatively here, $\rho = -0.48$), so we measure luminance on the eye region only. Sharpness on the whole eye crop is inflated by eyeglass frame edges, so iris quality is measured on a small iris-centred patch.

C. The Gaze Reliability Index

GRI is a no-reference quality index in the spirit of BRISQUE [15] and recent eye-tracking data-quality standards [16]. It judges the trustworthiness of the gaze signal from the signal itself, without ground truth. Let a clip have N sampled frames, of which N_{iris} contain detected iris landmarks and N_{valid} pass the geometric validity check. The five components are defined as follows.

Iris Detection Ratio. The share of frames in which the iris landmarks are found [20]:

$$\text{IDR} = N_{\text{iris}}/N. \quad (1)$$

Gaze Validity Ratio. A detected iris is geometrically valid when its centre lies between the inner and outer eye-corner landmarks of each eye [10]:

$$\text{GVR} = N_{\text{valid}}/N. \quad (2)$$

The missing rate $\text{MR} = 1 - \text{GVR}$ is reported as a diagnostic but is not part of the score, because it carries no information beyond GVR.

Temporal stability. Let c_t be the iris centre in pixel coordinates at valid frame t . Raw jitter is the mean displacement between consecutive valid frames [12]:

$$J = \frac{1}{M-1} \sum_{t=1}^{M-1} \|c_{t+1} - c_t\|_2, \quad (3)$$

$$\text{Jitter}_n = \min(1, J/J_{95}), \quad (4)$$

where M is the number of valid frames and J_{95} is the 95th percentile of raw jitter over the whole corpus (2.32 px at 10 FPS). The cap is set from the data rather than fixed in advance. Note that jitter is computed on the tracker's own output, so temporal instability measures the unreliability of the gaze estimate itself, not merely properties of the input image.

Eye openness. We use the standard two-pair eye aspect ratio [21]. With vertical landmark pairs (p_2, p_6) and (p_3, p_5) and a horizontal pair (p_1, p_4) for each eye,

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \|p_1 - p_4\|}, \quad (5)$$

$$\text{EAR}_n = \min(1, \overline{\text{EAR}}/0.4), \quad (6)$$

where $\overline{\text{EAR}}$ is the clip mean over both eyes and 0.4 is an open-eye baseline. Low EAR means the iris is partly hidden by the eyelid.

Iris image quality. Let P be a 25×25 pixel grayscale patch centred on the iris. Quality is the normalised variance of its Laplacian [15], [17]:

$$\text{IrisQ} = \min(1, \text{Var}(\nabla^2 P)/500). \quad (7)$$

Measuring quality on the iris patch, not the whole eye crop, keeps the score insensitive to eyeglass frame edges.

The final score is the equal-weighted mean of the five terms, each in $[0, 1]$:

$$\text{GRI} = \frac{1}{5} (\text{IDR} + \text{GVR} + (1 - \text{Jitter}_n) + \text{EAR}_n + \text{IrisQ}) \quad (8)$$

so that 1 denotes fully reliable gaze and lower values denote degraded signal; jitter enters as $1 - \text{Jitter}_n$ so that all five terms point in the same direction, larger meaning better. Equal weights are a deliberate, untuned baseline following the approach of [15]. We also record the mean eye-region luminance of each clip as a diagnostic, and divide clips into dark, neutral, and light tertiles from it.

D. Failure Taxonomy Detection

Clips in the lowest GRI decile are assigned to one or more of five categories. The thresholds are data-derived deciles, not preset constants. The categories are as follows: (1) morphological, where the clip mean EAR falls below the 10th percentile, reflecting a personal trait; (2) image quality, where IrisQ falls below the 10th percentile, reflecting an equipment-related factor; (3) low eye-region luminance, corresponding to the darkest tertile of eye brightness, which section IV-C shows to be a stable property of the individual; (4) sustained instability, where raw jitter exceeds the 90th percentile; and (5) detection failure, where IDR or GVR falls below 1. On a traced subset of 2,767 clips, covering all eyeglass wearers and 1,400 matched controls, we also annotate transient brightness-jump events at the frame level to identify the mechanism behind instability.

E. Statistical Analysis

Group comparisons use the Mann-Whitney U test with rank-biserial correlation as effect size, because at $n = 8,409$ a small p -value alone says little. Subject-level systematicity (RQ2) is measured with the intraclass correlation ICC(1), the share of GRI variance explained by subject identity, together with a χ^2 test of bottom-decile membership against a uniform allocation. Validation against labels (RQ3) uses χ^2 tests with Cramér’s V across GRI quartiles. Associations are reported with Spearman rank correlation. Human validation uses group-level comparison across GRI strata with the Mann-Whitney U test and the Spearman correlation between mean human rating and GRI. Together these analyses answer RQ1 on failure rates and drivers, RQ2 on systematicity, and RQ3 on whether GRI reflects real signal trustworthiness.

F. Human Validation

To test whether GRI reflects degradation that people actually perceive, and not only a numerical artifact, we conducted a blinded rating study. We sampled 60 clips stratified across three GRI levels, the bottom decile, a band around the median, and the top decile, with the low group further stratified across failure types so that darkness, low detection, blur, and eye-closure were each represented, and no subject contributed more than two clips. Clips were renamed neutrally and presented in random order. Three raters, blind to the GRI scores and to the purpose of the study, independently judged each clip on a three-point scale according to whether the iris and its gaze direction were followable for most, some, or little of the clip. The rater wording was fixed through a pilot before the main study.

IV. RESULTS

A. GRI Distribution and Detection Saturation

We first address RQ1. Across all 8,409 confirmed high-engagement clips, GRI has mean 0.727, standard deviation 0.087, and range 0.353 to 0.948. Figure 2 shows the distribution. Two observations matter. First, hard detection failure is nearly extinct: only 0.4% of clips show any frame in which the iris is not detected or fails the geometric check. On confirmed engaged students, the tracker essentially always returns a complete, confident gaze stream. Second, the quality behind that stream varies widely and continuously. There is no natural split between broken and working clips, only a gradient from clean to degraded signal. We therefore define low-GRI clips by the empirical bottom decile ($\text{GRI} < 0.62$, dashed line in Fig. 2) rather than by a preset threshold. Together these results answer the first half of RQ1: visible failure is rare, while silent quality degradation is common.

B. Failure Taxonomy

We next ask what drives the 841 bottom-decile clips. Figure 3 reports the share of low-GRI clips in each category; categories overlap, so shares sum to more than 100%. Sustained instability is the most common driver (62.4%), followed by low eye-region luminance (36.5%), low eye openness (25.8%),

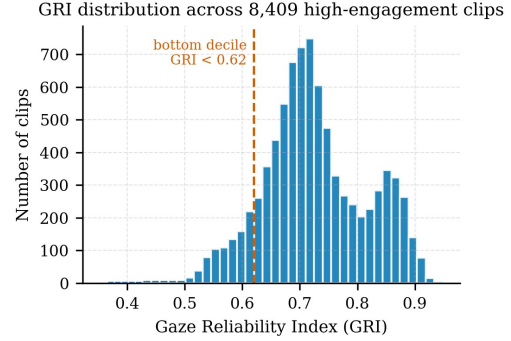


Fig. 2. GRI distribution across 8,409 confirmed high-engagement clips. Degradation is continuous; the sparse left tail extends to 0.35. Dashed line: the empirical bottom decile ($\text{GRI} < 0.62$).

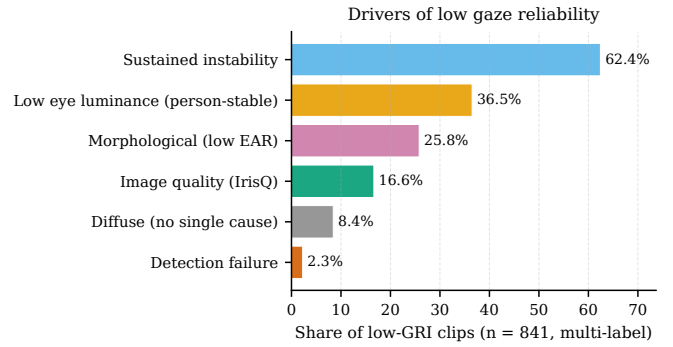


Fig. 3. Drivers of low gaze reliability among the 841 bottom-decile clips (multi-label, so shares sum to more than 100%; thresholds defined in Sec. III-D).

degraded iris image quality (16.6%), and hard detection failure (2.3%). Causes compound: 41.1% of low-GRI clips fall in two or more categories, and the largest overlap is between instability and low luminance (155 clips), which is physically plausible because a poorly lit iris is harder to localise stably. A further 8.4% of low-GRI clips show diffuse degradation, with several components mildly poor but none crossing its decile threshold. This is a property of a composite index: it accumulates mediocrity that no single signal check would flag. The taxonomy is not sensitive to the decile choice: at the worst 5% or 20% of clips, low luminance remains at 34–37% and the concentration on the same few subjects persists, with sustained instability growing more dominant toward the extreme tail. Figure 4 shows one representative frame per category with the iris landmarks overlaid.

C. Subject-Level Consistency

RQ2 asks whether degradation is random or systematic. It is systematic. Across the 110 subjects with at least five clips, the intraclass correlation is $\text{ICC}(1) = 0.710$, meaning 71% of all GRI variance is explained simply by which student is being recorded. This result is robust to label noise: restricting to clips

Representative low-GRI clips by failure category (iris landmarks in green)

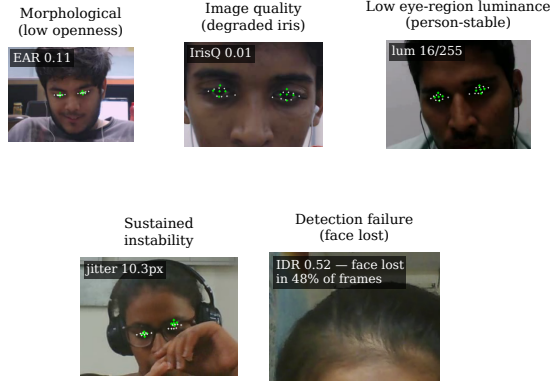


Fig. 4. One representative low-GRI clip per failure category; iris landmarks in green, the deciding measurement boxed. The detection panel shows a frame in which the face is lost.

with no elevated boredom, confusion, or frustration (5,929 clips) yields $ICC(1) = 0.726$. The most affected subject has mean GRI 0.586 with a standard deviation of only 0.035 across 120 clips, and 83% of those clips fall in the global bottom decile. The least affected subject averages 0.874. Ten subjects alone account for 36.7% of all bottom-decile clips, and a χ^2 test rejects a uniform allocation decisively ($\chi^2 = 1776$, $dof = 111$, $p < 10^{-290}$). Figure 5 plots each subject’s mean iris image quality against mean GRI. The two are strongly associated at the subject level (Spearman $\rho = 0.861$), while eye openness and eye-region luminance act mainly at the clip level through the taxonomy rather than through subject averages ($\rho = 0.125$ and 0.088 , both not significant). Eyeglass wearers (blue) are interleaved across the whole range, a first sign that glasses do not determine reliability; at the subject level, their mean GRI is 0.790 against 0.716 for non-wearers ($p = 0.001$), but with only 18 wearers and per-subject camera differences this difference should not be read causally; it more plausibly reflects per-subject differences in cameras and recording setups than any protective effect of glasses. Notably, all ten most affected subjects wear no glasses.

D. GRI Validation Against Engagement Labels

RQ3 asks whether GRI reflects real signal trustworthiness rather than being a purely technical score. We test this against the human labels themselves. Figure 6 shows both tests by GRI quartile. The share of clips labelled Very High rises monotonically from 39.1% in the lowest quartile to 54.7% in the highest ($\chi^2 = 108.9$, $p = 1.9 \times 10^{-23}$, Cramér’s $V = 0.114$), even though all clips come from confirmed engaged students. Annotators systematically rate students with degraded signal as less engaged. In addition, the share of clips carrying conflicting affect labels, meaning a confirmed engaged clip that is simultaneously labelled bored, confused, or frustrated at level two or above, falls from 36.2% in the lowest quartile to 24.3% in the highest ($\chi^2 = 75.2$, $p = 3.4 \times 10^{-16}$, $V = 0.095$). Both effects are small in magnitude, as expected

Reliability is a stable property of the student ($ICC = 0.71$)

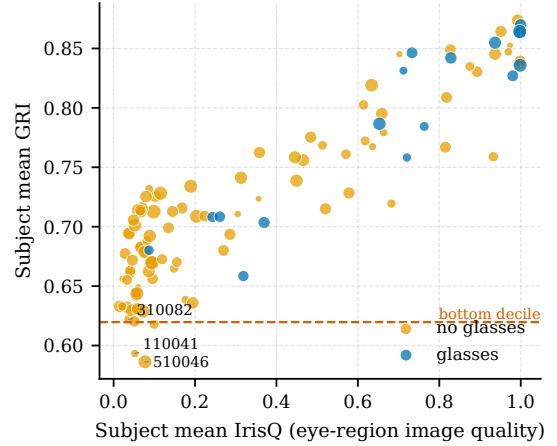


Fig. 5. One point per student (≥ 5 clips, $n = 110$); size reflects clip count. The same students sit low in every session ($ICC(1) = 0.71$). Dashed line: bottom-decile threshold. Eyeglass wearers (blue) are interleaved throughout.

since label noise has many sources, but both run monotonically in the same direction. Low GRI marks clips whose human judgments are weaker and less internally consistent, which is precisely what a trustworthiness measure should detect. The measurement disadvantage becomes a judgment disadvantage.

We further validate GRI against direct human perception. Three raters, blind to the scores, judged gaze visibility on 60 clips stratified across the GRI range (Section III-F). Mean ratings rise monotonically across the low, medium, and high GRI groups (1.9, 2.6, and 2.8 on the three-point scale), the low and high groups are clearly separated (Mann-Whitney, $p < 0.001$), and ratings correlate strongly with GRI (Spearman $\rho = 0.70$, $p < 0.001$), with 87% of rating pairs agreeing within one scale point. Human perception of gaze visibility therefore tracks GRI, confirming that the index captures degradation people recognise rather than a purely technical quantity.

E. Ablation

Figure 7 reports how much each component contributes to the GRI ranking, measured as one minus the Spearman correlation between the full index and the index with that component removed. Iris image quality contributes most (0.433), followed by temporal stability (0.182) and eye openness (0.024). Detection and validity contribute essentially zero, because both saturate at 1.0 for 99.6% of clips. We report this saturation as a finding rather than hiding it: on confirmed engaged students, modern iris tracking simply does not fail visibly, so all reliability information lives in quality and stability. We retain the two detection terms in the index because they are exactly the terms expected to matter in lower-quality capture settings than DAiSEE, such as uncontrolled home webcams, where the index is meant to generalise. The findings are also robust to the weighting itself: across 1,000 random weightings of the five components, the median index correlates $\rho = 0.90$ with the equal-weight GRI and yields $ICC(1) = 0.72$, with

Lower gaze quality → weaker, less consistent labels

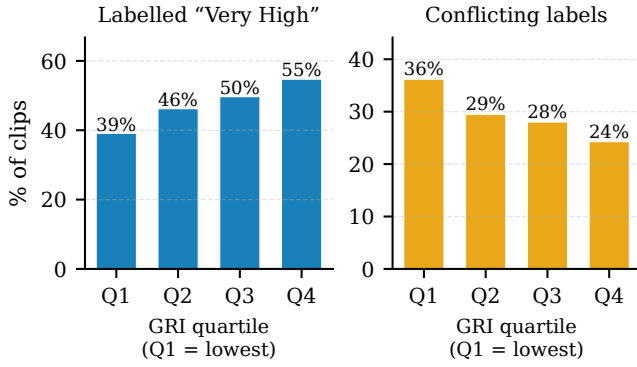


Fig. 6. Human labels degrade with gaze quality across GRI quartiles (Q1 lowest). Left: share graded Very High. Right: share with contradictory labels (engaged yet graded bored, confused, or frustrated ≥ 2), reflecting annotator uncertainty.

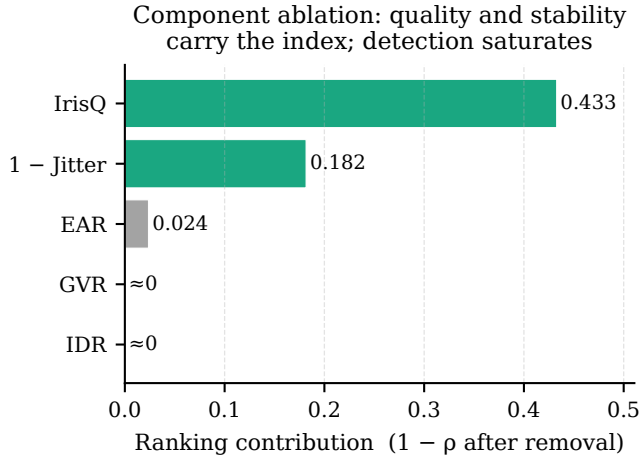


Fig. 7. Component ablation. Quality and stability carry the index, while detection terms saturate on engaged students and contribute no ranking information in this dataset.

degradation only under degenerate weightings dominated by the saturated detection terms, consistent with the ablation.

F. Downstream Impact

Finally, we quantify what these disparities mean for a system that acts on them. Consider a system that discards unreliable clips before assessing engagement, a natural use of any quality index. Discarding the bottom GRI decile removes 10% of clips overall, but it removes 45.3% of the data of the ten most affected students on average, and 83% of the data of the single most affected student. Clips in the darkest eye-luminance tertile are removed at 1.3 times the rate of the brightest tertile (11.0% against 8.6%). The burden of unreliability is therefore concentrated on specific individuals rather than spread thinly across the population, and a system that silently filters by quality would render those students nearly invisible to itself.

V. DISCUSSION

A. Reliability Implications

This study asked whether confirmed engagement guarantees reliable gaze extraction. It does not, but the failure mode is not the one usually assumed. The tracker almost never reports a problem. It returns a complete gaze stream for nearly every clip, while the quality of that stream spans from excellent to poor and the difference is invisible to any downstream consumer. A system built on raw webcam gaze therefore treats sharply unequal evidence as if it were equal, with no warning signal. The practical implication is direct: gaze should not enter an engagement model without a reliability estimate attached, and GRI provides one at no labelling cost. GRI is also more than an image-quality score: it incorporates the stability of the tracker’s own output, and it predicts the consistency of human judgments, which a measure of input sharpness alone has no reason to do.

B. Fairness Implications

The disparity is a stable property of the person, not the setting. Eye-region darkness is highly person-stable ($\text{ICC}(1) = 0.953$), yet it does not come from dark rooms: the always-dark subjects sit in scenes brighter than average while their eye regions image at less than half the brightness of other students. We hypothesise that skin tone, interacting with webcam auto-exposure that meters on the bright background, is one contributing factor, though we cannot test it: DAiSEE carries no demographic metadata, so we raise it as a fairness concern to verify with such data rather than a measured disparity. The pattern is consistent with documented camera biases against darker-skinned individuals [7], [9].

The popular suspect is also innocent: eyeglasses show no meaningful effect on any component, and none of the ten most affected students wears them. Frame-level inspection of the strongest brightness events found head-pose changes, hand occlusions, and auto-exposure shifts, not lens reflections. These transient events affect all students at a rate of roughly one per hundred frames and multiply instantaneous gaze displacement by about four, which carries an uncomfortable irony for engagement systems: a student who looks down to take notes is behaving exactly as an engaged student should, and at that moment the tracker silently emits corrupted gaze with full confidence.

One case study marks a limit of signal-based quality assessment. A student’s eyeglasses show strong screen reflection that intermittently occludes the iris to human inspection, yet the clip receives perfect detection, near-maximal iris sharpness (0.999), and above-average GRI (0.82). Per-clip aggregate scores register no trace of these occluded moments, and the reflection’s sharp edges may themselves inflate the sharpness measure. Frame-level visibility assessment, distinguishing a seen iris from a confidently estimated one, remains open.

We therefore recommend that gaze-based educational systems treat reliability as a matter of transparency and accountability, not only accuracy: report a per-student reliability

profile alongside any engagement estimate, disclose when and for whom signal quality is degraded, and never apply quality filtering silently, since Section IV-F shows it would disproportionately erase the students the system already serves worst. GRI provides the audit instrument for this at no labelling cost. GRI also enables a form of fairness intervention that does not require knowing students' sensitive attributes in advance: it acts on the observable degradation of the signal rather than on group identity, protecting affected students even when demographic labels are unavailable, as they are here.

C. Limitations and Future Work

Several limitations bound these findings and point to natural next steps. Pixel darkness is not a direct measure of skin tone, and DAiSEE provides no demographic metadata, so quantifying that component requires future work with demographic labels. The dataset's indoor lighting range is also narrow, leaving stronger lighting and genuine outdoor reflections untested, and only eighteen subjects wear glasses, which limits the power of the eyeglass comparisons. All results rely on a single tracker, MediaPipe Face Mesh, so other trackers may fail differently, and the transient-event analysis covers a traced subset rather than the full corpus. Beyond addressing these, the most promising direction is to make reliability actionable: rather than treating gaze as uniformly trustworthy, future systems could weight or gate the signal by its precomputed reliability, attenuating low-GRI input instead of discarding it, and in multimodal settings could shift toward complementary cues such as facial action units, head pose, posture, or audio when gaze reliability is low. This would turn per-student reliability from a passive diagnostic into an active fairness mechanism for gaze-based learning systems.

VI. CONCLUSION

We audited the reliability of webcam gaze on 8,409 confirmed high-engagement clips from 112 students and found that the question is not whether gaze extraction fails, but for whom its quality silently degrades. Detection succeeds almost always, yet 71% of the variation in signal quality is explained by student identity alone; the same students receive degraded signal in session after session, and human annotators rate exactly those students as less engaged. The drivers are person-stable eye-region darkness, image quality, and temporal instability, while eyeglasses, the commonly assumed culprit, show no meaningful effect. The Gaze Reliability Index makes this invisible disparity visible at no labelling cost, predicts which clips carry untrustworthy labels, and aligns with blinded human judgment of gaze visibility. Importantly, these patterns persist across multiple recording sessions and learning contexts, suggesting a systematic rather than incidental source of measurement bias. We argue that any gaze-based educational system should treat reliability as a first-class, per-student signal, because a system that cannot see how unevenly it sees will quietly fail the same students every time. Before educational AI can interpret where students look, it must first know whether it can reliably see them.

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